Final Report

Section 1: Meta Information

Project title: Movies vs Politics

Group ID: 27

Mentor Name: Avadh Hindocha

Name/Roll No.: 1) Mehakdeep/25b0918
2) Anisha/25b0921
3) Shreya/25b1045
4) Shrividya/25b1085

Honor Code and Discussions:
We did not consult or discuss our work with anyone outside our group.

LLM Usage Links: None

Link to Video Presentation:
https://drive.google.com/drive/folders/1qZExDgMkRBsTJTaNuHRUlXdph9BcOm4L

Github Repository Link:
https://github.com/TavanamShrividya/movies-vs-politics

Project tracking spreadsheet:
https://docs.google.com/spreadsheets/d/1liRBqMXGuKvWqg_8t2DW5kDkKSPTW-qSxFX5fM7xXvk/edit?gid=0#gid=0

Github handles and initial commit date:
1) mehakdeepk952-source; 3/11/2026
2) Anisha7366; 19/10/2026
3) shreyakhaire2; 03/11/2026
4) TavanamShrividya; 19/10/2026

Contribution Breakdown:
Equal contributions from all members.

1. Shrividya was responsible for finding all the required data sets
2. Anisha did the data cleaning
3. Mehakdeep and Shreya were responsible for generating various charts
4. Everybody contributed to calculations and report

Section 2: Technical Content

Core Problem Statement:

We plan to analyse movies over the past decades using their genres like action, thriller, romance, comedy, drama, horror, etc., against the political climate. We will be measuring the political climate and uncertainty with crime rates, protest numbers, and the margin of votes for the ruling party (to measure political dissatisfaction). We will also be focusing on standout politically tense times, such as wars.

Some questions we're looking to answer:
Does a politically unstable climate lead to more violent movies being made or lighter genres?
How was the political climate when romantic movies were on the rise?
What genres were most consumed at times when India was at war? Post - war?

We'll primarily be doing a decade-by-decade analysis except when the political party changed. We also plan to compare the characteristics of the movie genres against crime rates, protest numbers (protest type if required), war years, post-war years, and political party dissatisfaction. For this, we'll make graphs to look for patterns, and use the statistical methods we've learned in class to test the data. Our final graph is a line graph that correlates % of movies per genre each year vs a "political uncertainty coefficient" (defined later).

Our goal is to figure out which genres peaked during politically tense periods, what genres peaked during relaxed times, and how that differs over generations.

Data Set(s):

Dataset	Provenance	Scale	Structure	Accessibility
movies.csv	kaggle.com	74 * 9		free
elections.csv	GitHub repo-gitcheckoutnikhil	20 * 4		free
protest.csv	geopoliticaleconomist.com	23 * 2		free
crimes.csv	kaggle.com	13 * 2		free
wars.csv	mea.gov.in and jagranjosh (for military operations)	24 * 1		free

Tools and Software Packages: Pandas, Plotly

India Focus: The main dataset revolves around the top rated movies over the past decades in the Indian movie industry. This data is compared and plotted against the trends in Indian politics. This was achieved by defining and calculating a "political unrest coefficient" which depends on the crime rates, winning party vote percentage, protest and wars present in the year.

Challenges and Results Achieved: Technical hurdles, data cleaning roadblocks, key findings, and analytical results.
Technical Hurdles:

1. The data sets we initially planned to use turned out to be not free so we had to search again for free data sets.

2. While making graphs for comparing two types of data sets, one would be in points and other one would be in millions.

Data Cleaning roadblocks:
1. Raw movies dataset did not have valid movie years and missed genres for some data.
2. Some movie genres had very few movies each year, merged with similar genres/ dropped
3. Raw election data was candidate wise, dropping that to only focus on political party
4. Crime data was classified district wise, had to take sum state wise to get total statistics

Results

Define: Political Uncertainty Coefficient (PUC) is dependent on the ruling party's winning vote percentage, the number of crimes recorded in a given year, the number of protests held in that year, and the presence or absence of a war or major military operation.

- **Electoral Component (E□)** Random variable E□ is the normalized electoral component for the election year.

$$E_t = 1 - (V_t / 100)$$

where V□ represents the percentage of votes obtained by the ruling party in the general election closest to year t.

- **Crime Component (C□)** Random variable C□ is the normalized crime component for year t. Higher recorded crime counts are assumed to correspond to greater political uncertainty.

$$C_t = (Crimes_t - Crimes_{min}) / (Crimes_{max} - Crimes_{min})$$

where Crimes□ represents the number of recorded crimes in year t, and Crimes□ᵢ□ and Crimes□ₐₓ denote the minimum and maximum crime counts observed over the study period.

- **Protest Component (P□)** Random variable P□ is the normalized protest component for year t. A greater number of protests is assumed to correspond to greater political uncertainty.

$$P_t = (Protests_t - Protests_{min}) / (Protests_{max} - Protests_{min})$$

where Protests□ represents the number of protests recorded in year t, and Protests□ᵢ□ and Protests□ₐₓ denote the minimum and maximum protest counts observed over the study period.

- **Military Conflict Component (W□)** Random variable W□ is the military-conflict indicator for year t. It takes a value of 1 if a war or major military operation occurred during that year and 0 otherwise.

$$W_t = 1 \text{ if war/military operation occurred in that year; } 0 \text{ otherwise}$$

- **Construction of the PUC** The Political Uncertainty Coefficient is constructed as a weighted composite of the four components:

$$PUC_t = 100 \times (0.15E_t + 0.40C_t + 0.35P_t + 0.10W_t)$$

The weights assigned to the respective components are:

• Electoral component: 15%
• Crime component: 40%
• Protest component: 35%
• Military-conflict component: 10%

Through our project we found many key observations:

- There is a direct strong correlation between protests and non-fiction movies. It is as expected that people with more awareness about the history and present situations of the country would lead and participate in more protests and more protests provide more stories for the non-fiction genres.
- War reaction time is defined as the time it takes for the movie industry to react to wars and produce top rated war themed movies. This is calculated by taking the time difference between the year with a war-movie ratio spike and the previous most recent war. This gap decreases over time as expected which proves that the amount of time it takes people to know about a war, their motivation to care about such wars and the technology which affects the production time have improved over time.
- There is a negative correlation between crime movies and crime related movies, which was unexpected. There is however a positive correlation between comedy movies and crimes in India, which shows how the audience prefers lighter topics during uncertain times. There is a similar trend that can be observed with winning party vote percentage (lower % means more uncertainty) and comedy movies, which can be supported by a similar reasoning.
- Through linear regression, we found that protest numbers are the weakest predictors with average R^2 value 0.21 (in contrast to 0.55 of crime rates and 0.29 of election win percentage). We conclude that protest numbers explain genre trends less than crimes and elections do.
- Action movies have negative correlation with crime rates, election win percentage and protest numbers. The Audience seems to move away from action when political uncertainty rises. Drama is the opposite with positive slopes in all three, so Drama may be the comfort genre during political strain.
- There is a strong positive correlation between political uncertainty coefficient and comedy, drama movies. In contrast, greater political uncertainty leads to decrease in action and crime movies being made. We can conclude that people prefer lighter movies when things get uncertain.

Section 3: Frequently Answered Questions (FAQ)

Q.1) What are the key random variables in this project?

Answer: With such a wide range of data and so many ways of expressing the same political uncertainty we had to create a new random variable called Political uncertainty coefficient which was a weighted-sum of the individual factors we chose as measure of unrest. There are other random variables which are the fraction of movies in a certain genre for all genres over all years.

Q.2) How to calculate the political unrest coefficient?

Answer: Considering the factors that have extreme values (like wars/military operations that spike from 0 to 1) the weight of each factor was decided. The electoral component was given a considerably less weight even though it intuitively seems high because election choices are generally biased and a large number of citizens do not vote and if they do, very few are honest. As the ruling party's vote percentage decreases, political uncertainty is assumed to increase. This electoral component is also assumed to apply to neighboring non-election years. For a typical five-year election tenure, E_i is applicable for 2.5 years before and 2.5 years after the i-th election year.

Q.3) Is it true that crime-related movies have increased with crime rates over the years? Is it also true in the case of wars and war-related movies?

Answer: Interestingly there is a negative correlation between crime-related movies and crimes in India (correlation coefficient = -0.807). The reason behind this could be understood as follows. There is a notion in society that crime-related movies may be causing more crimes around the country and as a natural response the number of such movies being popularized decreases as society becomes more involved and aware of such changes. But the motivation behind crimes is not what people see on screen, it is rather the socioeconomic conditions that affect them. The unexpected trend can also be due to the fact that more people are willing and able to report crimes.

Q.4) Are there any unexpected trends you stumbled upon?

Answer: The correlation between crime-related movies and crimes was negative which was unexpected. There is also an unexpected positive correlation between crimes and comedy movies (correlation coefficient = 0.89), this can be understood by one of two reasonings either that people preferred comedy movies during times of high political unrest for the sake of comedic relief, or with a slight observation that comedy movies often involve “mocking” of political figures/ideas which made them even more popular among audiences.

Q5) Which political uncertainty factor proved to be the weakest predictor in linear regression?

Answer: Protest numbers seemed to be the weakest movie genre predictors (when estimated through R^2 values).